MIL-C-51355A(EA)
14 February 1979
SUPERSEDING
MIL-C-51355(MU)
18 May 1971

MILITARY SPECIFICATION

CARTRIDGE, 40 MM, CS, M651

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of cartridge.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Film, (Polyethylene Thin Gage).
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberbaord; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

FSC 1310

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Comment, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-B-131 - Barrier Material, Water Vaporproof, Flexible, Heat-Sealable.
MIL-B-46506 - Boxes, Ammunition Packing Wood, Wirebound.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 122-2-6 - Cartridge, 40 mm, CS, M651.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 122-2-6 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components shall conform to the specifications and drawings listed on TDPL 122-2-6 and subsidiary drawings.

3.2 Assembly. The cartridge shall be assembled as specified on Drawing D122-2-6.

3.3 Functioning. When tested as specified in 4.4.4.1 the cartridge shall function as follows:

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3.3.1 Velocity. The projectile shall have a corrected velocity of 242 ± 12 feet per second (fps) (73.76 ± 3.66 meters).

3.3.2 Penetration. The projectile shall penetrate the target.

3.3.3 Functioning following impact. The projectile shall function and disseminate CS within a maximum period of 5 seconds after target impact.

3.3.4 Burning time. The projectile shall have a burning time of 20 to 35 seconds after target impact. The burning time is defined as the time from first emission of smoke under pressure to cessation of emission of smoke under pressure.

3.3.5 Extraction. After firing, the cartridge case shall be extracted from the launcher by the spring operation of the extractor without auxiliary aid.

3.3.6 Cup seal. The disk formed by the coined area shall separate from the cup seal. The cup seal may split, but shall not separate from the cartridge case.

3.3.7 Premature burning. There shall be no premature burning (see 6.4) or evidence thereof in any of the test firing.

3.3.8 Misfires and duds. A misfire or a dud is considered a defect.

3.4 Preproduction. Prior to the start of regular production, a preproduction lot of cartridges shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship. The cartridge shall be free from contamination (oil, grease, CS or foreign matter) and damage such as dents, cracks and deformation which could affect it intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 200 cartridges shall be manufactured using the same methods, materials, equipment and processes as will be used during regular product-on.

4.3.2 Sampling.

4.3.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-414, level V.

4.3.3 Inspection procedure.

4.3.3.1 For examination. The sample cartridges shall be examined in accordance with the classification of defects (4.4.3.5) and MIL-STD-105.

4.3.3.2 For nondestructive test. Each cartridge in the lot shall be tested as specified in 4.4.4.2 (see 6.2). The acceptance number is zero.

4.3.3.3 For destructive test. The sample cartridges shall be tested in accordance with 4.4.4.1. Acceptability shall be in accordance with 4.4.3.3.

4.3.4 Acceptance/rejection criteria. The preproduction samples shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of cartridges, no more than one interfix number from one manufacturer of the following components is permitted:

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(a) Fuzes
(b) Body assemblies
(c) Case assemblies

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-414, level II.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample cartridges and the level A packaging shall be examined in accordance with the classification of defects (4.4.3.5) and MIL-STD-105.

4.4.3.2 For nondestructive test. Each cartridge in the lot shall be tested as specified in 4.4.4.2 (see 6.2). The acceptance number is zero.

4.4.3.3 For destructive tests.

(a) The sample cartridges shall be tested in accordance with 4.4.4.1. Acceptability with regard to velocity and burning time shall be determined by evaluation in accordance with MIL-STD-414, double specification limit, using an AQL of 2.5 percent defective for each variable.

(b) Acceptability with regard to parameters defined by 3.3.2, 3.3.3, 3.3.6, 3.3.7 and 3.3.8 shall be determined by evaluation in accordance with MIL-STD-105 using an AQL of 2.5 percent defective.

(c) The 2.5 AQL is to apply to these latter types of defects (penetration failure, failure to function within 5 seconds after impact, cup seal failure, premature burning, dud or misfire) collectively rather than individually.

(d) To determine the number of defects allowed, enter the appropriate table of MIL-STD-105 at the line on which the sample size most nearly matches the size of the sample actually tested. Any failure observed during the functioning test which the manufacturer can demonstrate to be the fault of a Government furnished component will not be considered a defect for purposes of determining acceptability of the lot under the provisions of this specification.

4.4.3.4 For critical defects. Each item in the lot shall be examined for the critical characteristics listed in the classification of defects.

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4.4.3.5 Classification of defects.

(a) Cartridge, 40 mm, CS, M651 (Drawing 122-2-6).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Head assembly not flush to cartridge case
102	Overall length incorrect
103	Surface of primer, flush tolerance incorrect
104	Torque incorrect
105	Sealant not removed from external surfaces
106	Workmanship (3.5)

(b) Case, cartridge (Drawing D122-2-35).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Eccentricity of 1.255 diameter incorrect
102	Eccentricity of 1.190 diameter incorrect
103	Eccentricity of 1.624 diameter to thread 1-1/8 - 18 NEF - 2B incorrect

(c) Head, case, cartridge (Drawing C122-2-37).

Categories	Defects
Critical:	
1	Holes (0.135) missing
Major:	AQL 1.0 percent defective
101	Eccentricity of thread 1-1/8 - 18 NEF - 2A to 0.665 diameter incorrect
102	Eccentricity of thread 0.775 - 24NS-2B incorrect

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4.4.3.5 Classification of defects (continued).

(d) Plug (Drawing C122-2-25).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Eccentricity of 0.119 diameter incorrect
102	Eccentricity of 0.1725 diameter incorrect

(e) Body (Drawing D122-2-64).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter 1.623 incorrect
102	Eccentricity of 1.623 diameter incorrect
103	Diameter 1.604 incorrect
104	Eccentricity of thread 1-15/32 - 24NS-2A incorrect

(f) Retainer (Drawing D122-2-4).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter 1.604 incorrect
102	Eccentricity of thread 1-15/32 - 24NS-2A incorrect

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4.4.3.5 Classification of defects (continued).

(g) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Packaging loose or incorrect
102	Component missing or incorrect
103	Container incorrect
104	Marking missing, illegible or incorrect
105	Strapping loose or missing
106	Packing container wood preservative not dry
107	Pallet quantity incorrect (see 5.4)

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Functioning.

(a) Immerse the complete cartridge under an eight inch (203.2 millimetres (mm)) head of water for 2 hours. The water temperature shall be within ± 5° F (2.8°C) of the ambient temperature. Remove the cartridge and wipe dry. The time lapse between the removal of the cartridge from the water and the functioning of the cartridge shall not exceed 1 hour. The cartridge shall be fired from an approved launching device (M79 launcher) at a target (ten by ten feet) located a distance of 125 feet (38.1 metres (m)) from point of firing. The target shall be constructed of 3/4 inch (19.05 mm) thick soft white pine or a double layer of 1/2 inch (12.7 mm) thick celotex. An approved backstop may be used to contain projectiles. Compute velocity as follows:

(b) Initial computation. The velocity shall be computed from a point 20 feet (6.1 m) from the launcher muzzle, from measurements made using two sets of screens, each set of screens being linked with an electronic counter. With one set of screens, the nearer screen shall be placed at 14 feet (4.3 m) from the muzzle, the other screen at 26 feet (7.9 m). With the other set of screens the nearer screen shall be placed at 15 feet (4.8 m) from the muzzle, and the other screen at 25 feet (7.6 m). The ratio of times obtained from the separate counters shall be 1.200 ± .005. Observations for functioning performance as well as velocity shall be made and recorded.

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(c) Corrected velocity. The mean velocity of the sample being tested shall be corrected by a factor determined for the day of test firing with a particular launcher. The correction factor to be applied to the test rounds shall be the arithmetic difference between the mean velocity of proof rounds on the day of test and the standard value (calibration value) of mean velocity given by the Government for the specific lot of proof rounds used. Each days test firing shall begin with the firing of at least two proof rounds. At least one more proof round will be fired after each group of 10 test rounds. At the end of each days firing at least two more proof rounds will be fired. The number of proof rounds to be fired during each day of testing shall be determined by table I. If the velocity value for a given proof round appears significantly different from the velocity values obtained for other proof rounds fired on that day, two additional proof rounds shall be fired in its stead. If the correction factor determined for application to the test rounds is greater than 15 feet (4.8 metres) per second, the test shall be considered invalid. A retest shall be conducted after an attempt is made to determine the cause of the large correction factor.

TABLE I. Standard deviation vs. proof rounds

Standard deviation given for proof round lot	Number of proof rounds to be fired
2.5 or less	24
2.6 to 3.0	35
3.1 to 3.5	47
3.6 to 4.0	61
4.1 to 4.5	78
4.6 and over	DO NOT USE

4.4.4.2 Cartridge diameter. The cartridge diameter shall be checked by using a gage (see 6.2(d)) having an inside diameter of 1.605 to 1.606 inches (40.767 to 40.792 mm) to ascertain if the cartridge will fit into the chamber of the launcher.

5. PACKAGING

5.1 Preservation and packaging, level A. Four bandoleers, containing six cartridges each, shall be packaged in a fiberboard box conforming to type V3c, style RSC of PPP-B-636, and having inside dimensions of approximately 11-7/8 by 10-1/2 by 4-5/8 inches (302 by 267 by 117 mm). Each

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bandoleer shall be separated by interlocking partitions fabricated of fiberboard conforming to type CF, class weather-resistant, grade W5c, of PPP-F-320. A printed 3 by 5 inch (76 by 127 mm) plastic instruction card, conforming to drawing C122-2-127, shall be placed with each folded bandoleer. Filler pads fabricated from the same material as the interlocking partitions shall be used as required to insure a tight pack. The box shall be closed in accordance with the requirements in the appendix of PPP-B-636. In addition all seams, corners and manufacturer's joint shall be taped in accordance with the appendix of the same specification for water-proofing of slotted style boxes. A barrier bag assembly consisting of a barrier bag and a bag liner shall be fabricated as follows: The barrier bag shall be a water-vaporproof, heat-sealable bag fabricated from barrier material conforming to class 1, of MIL-B-131. The approximate dimensions of the barrier material shall be 17-3/4 by 20-3/8 inches (451 by 5.8 mm). The bag liner shall be fabricated of 0.008 inch (0.203 mm) thick bag material conforming to type I, class 1, grade A, finish 1 of L-P-378. The approximate size of the bag liner shall be 16-1/2 by 20-3/8 inches (419 by 518 mm). The bag liner shall be fitted within the barrier bag. The fiberboard box shall be placed within the bag assembly and sealed in accordance with the heat sealing requirements of MIL-B-131.

5.2 Packing, level A. The cartridges packaged as specified in 5.1 shall be packed in an overseas type or class container having inside dimensions of approximately 12-7/8 long by 11-1/2 wide by 5-5/8 high (47 by 292 by 144 mm). A tolerance of ± 1/8 inch (± 3 mm) shall be used on the box dimensions. The container shall conform to PPP-B-601, style A, PPP-B-621 class 2, style 4, grade A or type 1, class 2 of MIL-B-46506. When the MIL-B-46506 container is used, a protective cover, Drawing C122-2-97, shall be placed over the unit package prior to placing it within the wirebound box. Fiberboard pads fabricated of the same material as the interlocking partitions, shall separate the contents from the top, bottom, sides and ends of the container. The container shall be closed and strapped in accordance with the requirements of the applicable container specification and its appendix, using galvanized strapping.

5.3 Marking. In addition to any special marking required by the contract, unit packages shall be marked in accordance with MIL-STD-129. The shipping container shall be marked in accordance with Drawing C122-2-128.

5.4 Palletization. When specified in the contract or order, 63 boxes packed as specified in 5.2, shall be palletized in accordance with Drawing C122-2-129 (see 6.2).

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6. NOTES

6.1 Intended use. The M651 cartridge is intended for use in the dispersal of CS agent in a specified target enclosure.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Palletization when required.
(c) Preproduction.

(1) Time allowed for submission of samples for Government test and evaluation after award of contract.
(2) Destination of samples and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

(d) The gage or gages (Picatinny Arsenal G60-12308827889) will be Government furnished.

6.3 Caution. Precautions should be taken to protect personnel and prevent fires or explosions incident to the preparation of the incendiary mixtures while assembling cartridges.

6.4 Premature burning. Premature burning is considered to have occurred if the round burns in the gun bore or in flight.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1310-A230

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-[illegible]A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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